Admissible level of deterministic noise that is orthogonal to the randomization on average over a Markov state

Working problem statement
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1. Context

We consider gradient-free minimization of a convex function f on a ball of radius R in $\mathbb{R}^d$ by the two-point method

$$\hat{g}(x) = \frac{d}{2\gamma}\big(O(x+\gamma e) - O(x-\gamma e)\big)\, e, \qquad e \sim \mathrm{Unif}(S^{d-1}),$$

where O is the function value with an error and γ is the smoothing radius.

Adversarial model. The oracle is $O(x) = f(x) + \delta(x)$ with $|\delta(x)| \le \Delta$; the error is fixed (a repeated query returns the same answer) and may depend on x arbitrarily. The noise then adds a *bias* of order $\sqrt{d}\,\Delta/\gamma$ to the gradient estimate, and accuracy ε requires $\Delta \lesssim \varepsilon\gamma/(\sqrt{d}R)$. The maximal admissible noise level in this model is now known up to logarithmic factors [2, 3]:

$$\Delta^{\mathrm{adv}}_{\max} \asymp \frac{\varepsilon^2}{\sqrt{d}\,MR} \qquad (M\text{-Lipschitz } f),$$

$$\Delta^{\mathrm{adv}}_{\max} \asymp \frac{\varepsilon^{3/2}}{\sqrt{d}\,\sqrt{L}\,R} \qquad (L\text{-smooth } f, \text{ Lipschitz with } M \sim \sqrt{L\varepsilon}).$$

Orthogonality to the randomization (survey [1], §4.4):

$$\mathbb{E}_e\big[\delta(x+\gamma e)\, e\big] = 0 \qquad \forall x. \tag{1}$$

Under (1) the noise causes no bias and enters only the variance of the estimate; heuristically the remaining condition is $d\Delta^2/\gamma^2 \lesssim M^2$, i.e. $\Delta \lesssim M\gamma/\sqrt{d}$ (for Lipschitz f with $\gamma \sim \varepsilon/M$ this is $\varepsilon/\sqrt{d}$, a factor MR/ε above the adversarial threshold). In practice, however, condition (1) almost never holds pointwise, at every x. The typical situation is that it holds *on average over some state* that changes over time.

Markovian noise. It is shown in [4] that for gradient-free methods with Markovian stochastic noise (mixing time τ, a chain that does not depend on the iterates) the cost of correlation is additive: the oracle complexity grows as $d + \tau$ rather than $d\tau$. A deterministic error is handled there only as a bounded adversarial one.

2. Noise class

Definition 1 (Noise orthogonal to the randomization on average over a Markov state)**.** The algorithm works in rounds $k = 1, 2, \ldots$. In each round the chain makes exactly one step, $s_{k+1} \sim P(s_k, \cdot)$, independently of the algorithm (exogenous chain; the initial law is arbitrary); the algorithm submits at most q points of the ball B_R ($q = 2$ below: both queries of the two-point estimate are made in the same state s_k), chosen from its internal randomness and all previous answers, and receives $O(y, s_k) = f(y) + \delta(y, s_k)$ for each; every round, with or without queries, consumes the budget. We assume:

(i) $|\delta(x, s)| \le \Delta$ for all x, s;

(ii) $\{s_k\}$ is a uniformly geometrically ergodic Markov chain with stationary law π and mixing time τ;

(iii) δ is jointly Borel in (x, s), and condition (1) holds under the stationary law for every radius $\gamma \in (0, \gamma_0]$:

$$\mathbb{E}_{s\sim\pi}\,\mathbb{E}_e\big[\delta(x+\gamma e, s)\, e\big] = 0 \qquad \text{for a.e. } x \in B_{R-\gamma} \text{ and all } \gamma \in (0, \gamma_0]; \tag{2}$$

equivalently, $\mathbb{E}_{s\sim\pi}\delta(x, s) = c$ for a.e. $x \in B_R$ and some constant c. For a fixed s the noise may be biased.

The class is denoted $\mathcal{N}(\Delta, \tau)$; it does not depend on γ, which becomes a parameter of the algorithm.

Fixed radius (remark). If (2) is required at one radius γ only (class $\mathcal{N}(\Delta,\tau,\gamma)$), the mean $m=\mathbb{E}_\pi\delta$ need not be constant: for bounded m on $\mathbb{R}^d$, $\widehat{m-c}$ is a single layer on the spheres $|\xi|=j_{d/2,k}/\gamma$ ("Bessel" noise; on B_R the class is larger). Such m acts as a systematic error at other radii (hypothesis H6). Two radii γ_1,γ_2 with $\gamma_1/\gamma_2\notin\{j_{d/2,k}/j_{d/2,l}\}$ already force $m\equiv c$ on $\mathbb{R}^d$ (on a ball this is cited, not re-proved), but a second radius above the first is quantitatively almost blind in high dimension.

Laziness. The main mechanism of slow mixing is the *laziness* of the chain: with probability α the state does not change at a step,

$$P_{\text{lazy}}=\alpha I+(1-\alpha)P,\qquad \text{mean holding time } \tfrac{1}{1-\alpha},\qquad \tau\gtrsim\tfrac{1}{1-\alpha}.$$

While a state s is held, the noise $\delta(\cdot,s)$ is the same, and its bias for fixed s repeats instead of averaging out: over $\sim\tau$ steps the deterministic error behaves like an adversarial one. Condition (2) starts to "work" only on the scale of the mixing time. In the limiting case $\alpha=1$ on part of the states (absorption) the chain does not mix at all.

Special cases.

- $\tau=1$ (independent states): the stochastic axiomatics of condition (1); for two-point methods of radius γ (class $\mathcal{A}_\gamma^{\text{2pt}}$ of §3), convex M-Lipschitz f, $\gamma\le\varepsilon/(4M)$, $\varepsilon<MR/8$ and $T\ge 1+16d^2M^2R^2/((d-1)\varepsilon^2)$, report H3 proves $\frac{\varepsilon\gamma\sqrt{T-1}}{4dR}\le\Delta_{\max}^{\mathbb{E},\text{2pt},\gamma}\le 32\pi\frac{\varepsilon\gamma\sqrt{T}}{dR}$, i.e. the threshold $\Theta(\varepsilon\gamma\sqrt{T}/(dR))$; in both bounds Δ may be replaced by the oscillation $\omega_\delta(2\gamma)/2=\frac12\sup_s\sup_{\|y-z\|\le2\gamma}|\delta(y,s)-\delta(z,s)|$;
- an external Markov data chain: the model of [4] with an additional deterministic component;
- *rounding* of function values to the grid $u\mathbb{Z}$: the state is the position of $f(x)$ on the grid (the offset). It is moved both by the data and by the method itself: on a step where the two rounded values coincide (a "tie") the point does not move, and with the same data the offset repeats. This is a lazy chain with holding probability $\alpha=P_{\text{tie}}(\omega,r)=2\Phi\big((\frac12-|\omega|)/r\big)-1$, where ω is the offset and $r=\gamma\|\nabla f\|/u$; the kernel of the chain depends on the iterates, and for $r\ll1$ and unchanged data the states in the middle of a cell are practically absorbing. For this case condition (2) has been verified exactly: for f affine on the stencil and Gaussian directions, $\mathbb{E}_e\hat g=\kappa(c,r)\nabla f$, where κ is a wrapped Gaussian of the offset c with width $r=\gamma\|\nabla f\|/u$, and $\int_0^1\kappa\,dc=1$. Directly: with a uniform offset, $\mathbb{E}_c\delta$ is the period mean of the rounding sawtooth, a constant, so (2) holds for all radii. The rounding chain, however, is driven by the method, which the exogenous-chain protocol above excludes; rounding belongs to $\mathcal{N}$ only after an extension to method-induced chains (hypothesis H8).

3. Problem

Goal

For the class $\mathcal{N}(\Delta,\tau)$ and the classes of Lipschitz and smooth convex functions, determine, as a function of the budget T (number of rounds of the protocol of §2),

$$\Delta_{\max}(\varepsilon,\beta;d,\tau;T)=\sup\big\{\Delta:\ \exists\,\mathcal{A}\text{ with }T\text{ rounds}\ \ \forall f\ \ \forall\delta\in\mathcal{N}(\Delta,\tau):\ \ \mathbb{P}\big(f(\hat x_{\mathcal{A}})-f^*\le\varepsilon\big)\ge1-\beta\big\},$$

and its expectation variant $\Delta_{\max}^{\mathbb{E}}(\varepsilon;d,\tau;T)$, with the requirement $\mathbb{E}f(\hat x_{\mathcal{A}})-f^*\le\varepsilon$; by Markov's inequality $\Delta_{\max}(\varepsilon,\beta;\cdot)\ge\Delta_{\max}^{\mathbb{E}}(\beta\varepsilon;\cdot)$. The *complexity-preserving level* is $\Delta_{\text{cp}}(C)=\Delta_{\max}$ at $T=C\,dM^2R^2/\varepsilon^2$. For two-point methods of radius γ we write $\Delta_{\max}^{\text{2pt},\gamma}$: the supremum is taken over the class $\mathcal{A}_\gamma^{\text{2pt}}$ of algorithms that in round t draw a direction $e_t\in S^{d-1}$ from a law fixed in advance (it may depend on t and on the internal randomness, not on previous answers), choose a centre y_t and a radius $\gamma_t\in(0,\gamma]$ from all previous answers, and observe only the difference $O(y_t+\gamma_te_t,s_t)-O(y_t-\gamma_te_t,s_t)$ (report H3, Def. 2.2; the class contains ZO-SGD with any step rule, batching, averaging and restarts). Whether $\Delta_{\max}$ is meant for this class or for all algorithms is open: under the conditions of report H3, for $\mathcal{A}_\gamma^{\text{2pt}}$ the case $\tau=1$ is settled up to constants, while for all algorithms it is only bracketed. Without a bound on T (in particular with a polynomial number of oracle calls) $\Delta_{\max}=\infty$ (report H1: for this class, and for the fixed-radius class with $\gamma\le\varepsilon/(4M)$). Obtain two-sided bounds on $\Delta_{\max}(T)$:

- **upper bound**: a construction of a hard function and an admissible noise for which no algorithm reaches accuracy ε (the existence side, following the scheme of [2]);
- **lower bound**: a two-point method with explicit parameters (radius, batch, step) that reaches ε for the given Δ.

Compare the resulting thresholds with the adversarial $\Delta_{\max}^{\text{adv}}$.

Expected picture.

$$\Delta_{\max}^{\text{adv}}\ \lesssim\ \Delta_{\text{cp}}(C)=\Delta_{\max}\big(T=C\,dM^2R^2/\varepsilon^2\big)\ \lesssim\ \frac{M\gamma}{\sqrt{d}}.$$

The lower side is proven in report H1 in the expectation variant: $\Delta_{\mathrm{cp}}(C) \geq M\gamma/\sqrt{2d+8\tau-3}$ for $\gamma \leq \varepsilon/(4M)$ ($\gamma < \varepsilon/(4M)$ in the a.e. reading of (2)) and $C \geq \frac{149}{4}(6+t_0/d)$, $t_0 = \tau\lceil\log_2(16MR/\varepsilon)\rceil$; e.g. $C = 261$ if $t_0 \leq d$. For $\tau \gg d$ the required C grows like $\tau\log(MR/\varepsilon)/d$. For independent states ($\tau = 1$), convex M-Lipschitz f, $\gamma \leq \varepsilon/(4M)$ and $\varepsilon < MR/8$, report H3 gives $\Delta_{\mathrm{cp}}^{\mathbb{E}}(24) \geq 1.034\, M\gamma/\sqrt{d}$ (expectation variant; ZO-SGD with one idle round, no burn-in). The upper side is proven in report H3 for two-point methods of radius γ (the class $\mathcal{A}_\gamma^{\mathrm{2pt}}$ defined above): $\Delta_{\mathrm{cp}}^{\mathbb{E},\mathrm{2pt},\gamma}(C) \leq 32\pi\sqrt{C}\, M\gamma/\sqrt{d}$ for $\varepsilon < MR/8$; the hard instance (linear f, rough cell noise with density $\cos^2(\pi t/2)$) has independent states and lies in $\mathcal{N}(\Delta,\tau)$ for every τ. For arbitrary algorithms the upper side is open. While the chain mixes, the threshold is determined by the variance and deteriorates as τ grows, i.e. as the laziness α grows: the bias accumulates over the holding time. When mixing breaks down ($\alpha \to 1$, absorbing states), the adversarial regime returns. A separate question is how the threshold depends on laziness generated by the method itself (rounding) rather than by an external chain.

Plan of work.

1. $\tau = 1$: rewrite the bound of §4.4 [1] in terms of the variance; construct a hard function for the class with condition (1) in stochastic form. (Done in report H3 for two-point methods of radius γ with direction laws fixed in advance; open for adaptive direction laws and for arbitrary algorithms.)
2. External chain, $\tau > 1$: transfer the technique of [4] (a direction per sample, multilevel Monte Carlo) to the component that is biased for a fixed state; obtain the dependence of the threshold on τ.
3. Lazy chain that depends on the iterates (rounding): relation between laziness and the tie probability, mixing conditions, absorbing states, comparison with the adversarial threshold.

Notation. The smoothing radius is γ here and μ in the experiments of this repository; they denote the same quantity.

References

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